KUET · ESE 3107 · Solar Thermal Engineering

ESE 3107 Question Map

Every question from the 2019, 2020, 2021, 2022, 2023 term papers, grouped by topic and ranked by how often it comes back and how many marks it carries. Answers are in the ESE 3107 Study Pack (HTML and PDF) next to this file.

24-hour list | Pass plan | Frequency grid | Section A | Section B

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 2 h for Part A and 3 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2023 ×5, 2022 ×4, 2021 ×3, 2020 ×2, 2019 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Retake tip: read the Concept Primer first (30 minutes). Most marks here are theory you can learn by heart: collectors (A5), radiation laws (A3), instruments and radiation components (A2), concentrators (B2) and applications (B4). For numericals, the sun-angle calculator covers A1 and the storage-tank and still calculators cover B3 and B4. FPC performance (B1) and radiation networks (A4) are the hardest: leave them for last.

Section A: The sun, radiation heat transfer and collectors

Block 1: very likely and easy · 1.2 h

Topic	Chance	Level	Covered
A5 Solar collectors: classification, flat plate construction, materials, glazing, evacuated tube and heat pipe	Very likely 100%	Easy	26%
A2 Solar radiation: extraterrestrial vs terrestrial, components, clearness index, measuring instruments, tilted surfaces	Very likely 100%	Easy	46%

A3 Radiation laws: blackbody, Planck, Stefan-Boltzmann, Wien, Kirchhoff, gray body, intensity	Very likely 100%	Easy	61%

Block 2: very likely, medium effort · 0.8 h

Topic	Chance	Level	Covered
A1 Sun-earth geometry: angles, solar time, sunshine hours, angle of incidence	Very likely 100%	Medium	79%

✔ Pass line: 79% of Section A marks covered, enough to choose 3 of the 4 questions (about 2.0 h)

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
A4 View factors and radiation exchange: relations, networks, reradiating surface, shields	Very likely 100%	Hard	100%

Section B: Collector performance and applications

Block 1: very likely and easy · 0.8 h

Topic	Chance	Level	Covered
B4 Other applications: drying, desalination (solar still), cooling, pool vs pond, solar thermal power	Very likely 100%	Easy	32%
B2 Concentrating collectors: definitions, classification, maximum concentration, tracking, materials	Very likely 100%	Easy	51%

Block 2: very likely, medium effort · 0.8 h

Topic	Chance	Level	Covered
B3 Solar water heating and storage: thermosiphon, ICS, direct/indirect, heat pump, storage tank numericals	Very likely 100%	Medium	71%

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
B1 Flat plate collector performance: losses, U_L, F', F_R, critical radiation level, testing, numericals	Very likely 100%	Hard	100%

✔ Pass line: 100% of Section B marks covered, enough to choose 3 of the 4 questions (about 2.6 h)

Pass plan

Five papers (2019-2023). Normal papers: 210 marks, answer any THREE of four questions in each section (35 marks each). 2020 was a short exam (120 marks, any TWO of three per section, 30 marks each). Section A (Q1-Q4) = the sun and its radiation (angles, solar time, instruments, tilted surfaces), radiation heat transfer (laws, view factors, networks) and collector hardware (flat plate, evacuated tube). Section B (Q5-Q8) = flat-plate performance (losses, U_L, F', F_R), concentrating collectors, water heating and storage, and other applications (drying, stills, cooling, ponds, power). **Minimum pass set:** Section A: Q1 (definitions + a sun-angle numerical), Q2 (clearness index, instruments, blackbody laws), Q4 (collectors: FPC parts, ETC, heat pipe). Section B: Q6 (concentrators: definitions, classification, tracking, C_{max}), Q7 (water heating + storage-tank numerical), Q8 (drying, still numerical, pond, central receiver). Answers follow the teachers' notes (Khaja sir lectures 1-6, Hasan sir's collector, concentrator and application notes and CT2 notes), which follow Duffie & Beckman, Kalogirou, G. N. Tiwari and Cengel; every numerical was re-checked in Python.

SECTION A: THE SUN, RADIATION HEAT TRANSFER AND COLLECTORS

4 topics

cover 79% of all Section A marks asked

- Q1: definitions, sunshine-hour derivation, solar time / incidence angle numerical
- Q2: clearness index, instruments, tilted surface, blackbody, E = πI, Kirchhoff
- Q4: collector classification, FPC parts, glazing, ETC and heat pipe
- Backup Q3: view factor rules, parallel plates, reradiating network, shields

SECTION B: COLLECTOR PERFORMANCE AND APPLICATIONS

3 topics

cover 71% of all Section B marks asked

- Q6: concentrator definitions, classification, tracking, C_{max}
- Q7: water heating systems + storage-tank numerical
- Q8: drying, still numerical, pool vs pond, central receiver
- Backup Q5: FPC losses, critical radiation level, F', U_L

- **Section A, pick Q1, Q2, Q4:** Q1 = definitions (air mass, hour angle, zenith, azimuth, solar constant) + sunshine-hour derivation + a sun-angle numerical (A1, A2); Q2 = clearness index/diffuse fraction, tilted surface, blackbody and E = πI, Stefan-

Boltzmann, Kirchhoff (A2, A3); Q4 = solar collector classification, FPC construction, glazing, ETC and heat pipe (A5). Q3 (view factors and networks, A4) is the backup.

- **Section B, pick Q6, Q7, Q8:** Q6 = concentrator definitions, classification, tracking, C_{max} = 46,747 (B2); Q7 = thermosiphon/direct/indirect/ICS + storage-tank numerical (B3); Q8 = drying, solar still numerical, pool vs pond, central receiver/dish (B4). Q5 (FPC losses, U_L, F′, B1) is the backup.
- **Repeated numericals:** sunshine hours at Khulna (2021, also derivation 2022); storage tank hour by hour (2019, 2022); solar still distillate (2020, 2021); U_L of an FPC (2019, 2021, 2022); parallel-plate radiation and view factors (every year).
- **Repeated theory:** $E_b = \pi I_b$ (2×), parallel plates by network (2×), three-zone reradiating network (2×), pyranometer/pyrheliometer (4×), FPC construction (4×), heat-pipe ETC (3×), pool vs pond (2×), central receiver (3×).
- **Always draw:** the sun-angle sketch, FPC cross-section, ETC/heat pipe, thermosiphon loop, concentrator types, solar still, the radiation network.

Frequency grid

Marks asked from each topic in each paper. Darker means more marks. Click a topic to jump to its questions.

Marks per paper: 1–8 9–16 17–24 25+

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION A: THE SUN, RADIATION HEAT TRANSFER AND COLLECTORS: MARKS ASKED PE

Topic	2019	2020	2021	2022	2023	Years	Marks	
A5 Solar collectors: classification, flat plate construction, materials, glazing, evacuated tube and heat pipe	24	28	41	35	45	5/5	173	Pass se
A2 Solar radiation: extraterrestrial vs terrestrial, components, clearness index, measuring instruments, tilted surfaces	27	18	32	26	30	5/5	133	Pass se
A3 Radiation laws: blackbody, Planck, Stefan-Boltzmann, Wien,	27	12	15	25	20	5/5	99	Pass se

Topic	2019	2020	2021	2022	2023	Years	Marks	
Kirchhoff, gray body, intensity								
A1 Sun-earth geometry: angles, solar time, sunshine hours, angle of incidence	25	20	23	25	28	5/5	121	Pass set
A4 View factors and radiation exchange: relations, networks, reradiating surface, shields	26	20	35	29	27	5/5	137	Next to a

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION B: COLLECTOR PERFORMANCE AND APPLICATIONS: MARKS ASKED PER PAPE

Topic	2019	2020	2021	2022	2023	Years	Marks	
B4 Other applications: drying, desalination (solar still), cooling, pool vs pond, solar thermal power	35	30	52	35	52	5/5	204	Pass se
B2 Concentrating collectors: definitions, classification, maximum concentration, tracking, materials	35	10	25	25	28	5/5	123	Pass se
B3 Solar water heating and storage: thermosiphon, ICS, direct/indirect, heat pump, storage tank numericals	35	20	18	35	18	5/5	126	Pass set
B1 Flat plate collector performance: losses, U_L, F', F_R, critical radiation level, testing, numericals	46	22	39	45	32	5/5	184	Next to a

Section A: The sun, radiation heat transfer and collectors

questions 1-4

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

A5 Solar collectors: classification, flat plate construction, materials, glazing, evacuated tube and heat pipe

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 173 marks in total

STUDY THESE

- Collector = heat exchanger turning sunlight into heat of a fluid; stationary (FPC, ETC, CPC) vs concentrating; low/medium/high temperature
- FPC parts: glazing, absorber plate with selective coating, riser tubes and headers, insulation, casing; heat transfer in an FPC
- Glazing: low-iron glass transmits short-wave (0.3-2.5 µm), opaque to long-wave (greenhouse effect); absorber: copper/aluminium, black chrome
- Evacuated tube: vacuum removes convection/conduction losses; heat pipe ETC; better at high temperature and in cold/cloudy climates

> 2023 Q4(a) 6 marks
>
> What is a solar collector? Classify by temperature range.

> 2023 Q4(b) 10 marks
>
> Working principle of an FPC with schematic.

> 2023 Q4(c) 10 marks
>
> Materials for components of (i) FPC, (ii) concentrating collector.

> 2023 Q4(d) 9 marks
>
> “ETCs perform better than FPCs”: justify.

> 2023 Q6(b) 10 marks
>
> Construction of a heat pipe ETC with diagram.

> 2022 Q4(a) 10 marks
>
> What is a solar thermal collector? Classify solar collectors.

> 2022 Q4(c) 10 marks
>
> Components of an FPC with a neat sketch.

> 2022 Q4(d) 5 marks
>
> Function of glazing materials.

> 2022 Q4(b) 10 marks
>
> Working principle of an evacuated tube heat-pipe collector.

> 2021 Q4(a) 13 marks
>
> Construction details of an FPC with schematic.

> 2021 Q5(b) 6 marks
>
> Compare materials used for glazing and absorber.

> 2021 Q4(b) 10 marks
>
> Advantages and limitations of ETCs.

> 2021 Q4(c) 12 marks
>
> Working principle of an ETC with a heat pipe.

> 2020 Q3(b) 10 marks
>
> Construction details of a typical FPC with schematic.

> 2020 Q4(b) 8 marks
>
> Short note on glazing material.

> 2020 Q3(c) 10 marks
>
> Working principle of an evacuated tube collector; advantages.

> 2019 Q4(a) 9 marks
>
> What is a solar collector? Heat transfer mechanism in a flat plate collector.

> 2019 Q4(c) 10 marks
>
> What is an ETC? Working principle.

> 2019 Q5(a) 5 marks
>
> Why is an ETC preferable over an FPC?

Full answer: ESE 3107 Study Pack, topic A5.

A2 Solar radiation: extraterrestrial vs terrestrial, components, clearness index, measuring instruments, tilted surfaces

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 133 marks in total

STUDY THESE

- Solar constant G_{sc} = 1367 W/m^2; extraterrestrial (above the atmosphere) vs terrestrial (at the ground, after absorption and scattering); air mass m = 1/cosθ_z

- Beam, diffuse, global (total) radiation; clearness index $K_T = H/H_o$; diffuse fraction H_d/H falls as K_T rises (Erbs, Liu-Jordan)
- Instruments: pyranometer (global/diffuse, with shading ring), pyrheliometer (beam, tracks the sun), sunshine recorder (Campbell-Stokes)
- Tilted surface: $I_T = I_b R_b + I_d(1 + \cos\beta)/2 + I\rho_g(1 - \cos\beta)/2$; monthly $\bar{H}_T = \bar{R}\bar{H}$ (Liu-Jordan)

> `2023` `Q1(d)` `7 marks`
>
> Differentiate between terrestrial and extraterrestrial radiation.

> `2023` `Q2(a)` `7 marks`
>
> Define beam, global radiation and diffuse fraction; daily diffuse fraction vs clearness index.

> `2023` `Q3(d)` `8 marks`
>
> What is geometric factor? Monthly average daily radiation on tilted surfaces.

> `2023` `Q2(d)` `8 marks`
>
> Monthly average hourly clearness index; overall tilt factor.

> `2022` `Q2(a)` `8 marks`
>
> What is diffuse fraction? How is daily diffuse fraction related with daily clearness index?

> `2022` `Q1(c)` `10 marks`
>
> Working principle of a pyrheliometer.

> `2022` `Q2(b)` `8 marks`
>
> What is geometric factor? Monthly average daily radiation on tilted surfaces.

> `2021` `Q1(c)` `6 marks`
>
> Distinguish between extraterrestrial and terrestrial radiation.

> `2021` `Q1(e)` `6 marks`
>
> What is meant by clearness index and diffuse fraction?

> `2021` `Q2(a)` `12 marks`
>
> Working principle of pyranometer; how can beam radiation be measured with it?

> `2021` `Q2(b)` `8 marks`
>
> Incident radiation on a tilted surface; beam radiation on a tilted surface using the geometric factor.

> `2020` `Q1(b)` `10 marks`
>
> Working principle of pyranometer with schematic diagram.

2020 Q2(a) 8 marks

How to calculate solar radiation on a tilted surface?

2019 Q2(a) 8 marks

Clearness index and diffuse fraction; how is diffuse fraction calculated from clearness index?

2019 Q2(b) 9 marks

Define: geometric factor, beam radiation, extraterrestrial radiation.

2019 Q1(b) 10 marks

What is a pyrheliometer? Working principle.

Full answer: ESE 3107 Study Pack, topic A2.

A3 Radiation laws: blackbody, Planck, Stefan-Boltzmann, Wien, Kirchhoff, gray body, intensity

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 99 marks in total

STUDY THESE

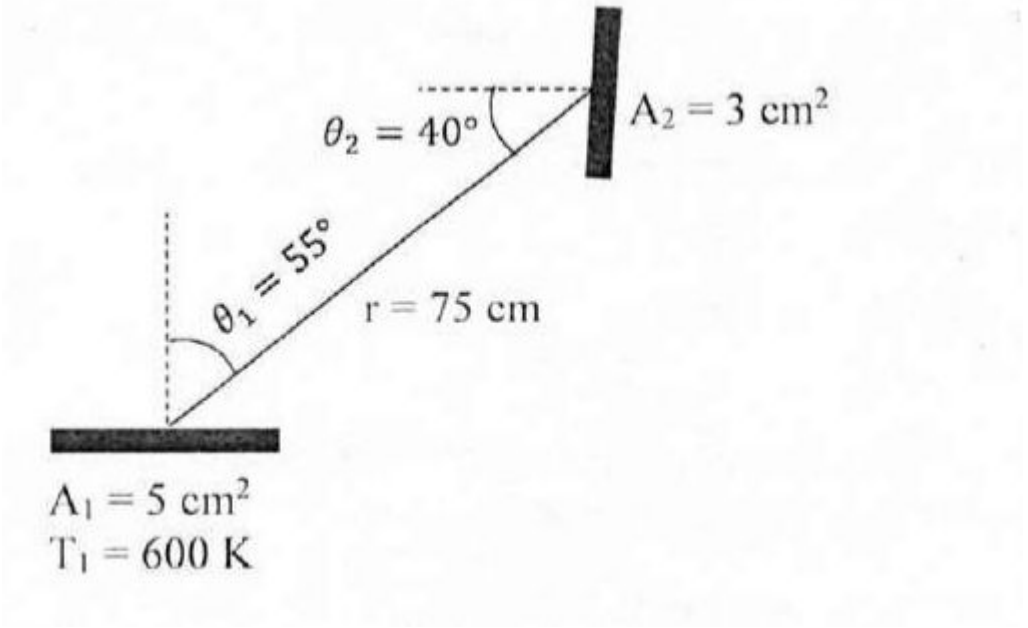
- Blackbody: absorbs all incident radiation, maximum emitter at every T and λ, diffuse emitter
- Planck $E_{b\lambda} = C_1/[\lambda^5(e^{C_2/\lambda T} - 1)]$; Stefan-Boltzmann $E_b = \sigma T^4$; Wien $\lambda_{max}T = 2898\ \mu m\ K$
- Emissivity, gray body (ε independent of λ); Kirchhoff ε = α at the same T; radiosity, irradiation
- Intensity I and solid angle $d\omega = dA\cos\theta/r^2$; diffuse emitter $E = \pi I$

2023 Q2(b) 8 marks

What is a blackbody? Show that $E_b = \pi I_b$.

2023 Q2(c) 12 marks

$A_1 = 5\ cm^2$ blackbody at 600 K, $A_2 = 3\ cm^2$ at r = 75 cm, $\theta_1 = 55°$, $\theta_2 = 40°$: spectral emissive power at 3 µm, solid angle, radiation striking A_2.



`2022 Q2(c) 9 marks`

Black body; state and explain the Stefan-Boltzmann law.

`2022 Q3(c) 6 marks`

Explain Kirchhoff's law of radiation.

`2022 Q2(d) 10 marks`

Define: emissive power, emissivity, gray body, radiosity, summation rule of view factor.

`2021 Q2(c) 9 marks`

Black body; state and explain the Stefan-Boltzmann law.

`2021 Q2(d) 6 marks`

State Kirchhoff's law.

`2020 Q2(b) 12 marks`

Features of a blackbody; show that $E_b = \pi I_b$.

`2019 Q2(c) 10 marks`

Explain the Stefan-Boltzmann law and Wien's displacement law.

`2019 Q2(d) 8 marks`

What is a gray body? Explain Kirchhoff's law.

`2019 Q3(b) 9 marks`

Define: radiosity, reradiating surface, irradiation.

Full answer: ESE 3107 Study Pack, topic A3.

A1 Sun-earth geometry: angles, solar time, sunshine hours, angle of incidence

Pass set · Very likely · 100% · Medium

Asked in 5 of 5 papers · 121 marks in total

STUDY THESE

- Latitude ϕ, longitude, declination $\delta = 23.45 \sin[360(284 + n)/365]$, hour angle $\omega = 15(ST - 12)$
- Zenith θ_z, altitude $\alpha = 90° - \theta_z$, solar azimuth γ_s, surface azimuth γ, slope β, angle of incidence θ
- Solar time = standard time + $4(L_{loc} - L_{st})$ + E (east longitudes); equation of time E
- Sunset hour angle $\omega_s = \cos^{-1}(-\tan\phi \tan\delta)$; day length $N = (2/15)\omega_s$; geometric factor $R_b = \cos\theta/\cos\theta_z$

> 2023 Q1(a) 10 marks
>
> Define: solar constant, azimuth angle, irradiation, air mass.

> 2023 Q1(c) 6 marks
>
> What is sunshine hour? How is it related to declination? Explain.

> 2023 Q1(b) 12 marks
>
> Angle of incidence on a surface at Nirala (22.8°N), 11:00 solar time, February 15, tilted 25°, 8° west of south; geometric factor.

> 2022 Q1(a) 15 marks
>
> Define: air mass, hour angle, solar azimuth angle, beam radiation, zenith angle.

> 2022 Q1(b) 10 marks
>
> Derive $N = (2/15)\cos^{-1}(-\tan\phi \tan\delta)$.

> 2021 Q1(b) 9 marks
>
> Define with schematic diagram: zenith angle, solar altitude angle, solar azimuth angle.

> 2021 Q1(a) 7 marks
>
> What is sunshine hour? How is it related to declination?

> 2021 Q1(d) 7 marks
>
> Sunshine hours for Khulna (22.80°N, 89.24°E) on January 18.

> 2020 Q1(a) 9 marks

Define: sunshine hour, clearness index, hour angle.

2020 Q1(c) 11 marks

Solar time corresponding to 11 AM on March 8 at Khulna (89.54°E).

2019 Q1(a) 15 marks

Define: solar constant, air mass, solar time, hour angle, surface azimuth angle.

2019 Q1(c) 10 marks

60° sloped surface facing 25° west of south, 3:00 PM solar time, March 5, latitude 45°: sunrise, solar altitude and azimuth; sunrise and sunset on the surface.

Full answer: ESE 3107 Study Pack, topic A1.

A4 View factors and radiation exchange: relations, networks, reradiating surface, shields

Next to add · Very likely · 100% · Hard

Asked in 5 of 5 papers · 137 marks in total

STUDY THESE

- View factor F_{ij} = fraction of radiation leaving i that strikes j directly; reciprocity $A_iF_{ij} = A_jF_{ji}$; summation $\Sigma F_{ij} = 1$; superposition; crossed strings
- Surface resistance $(1 - \varepsilon)/(\varepsilon A)$, space resistance $1/(A_iF_{ij})$; parallel plates $q = \sigma(T_1^4 - T_2^4)/(1/\varepsilon_1 + 1/\varepsilon_2 - 1)$
- Reradiating (insulated) surface: net radiation 0, $J = E_b$; three-zone network
- Radiation shields: N shields with equal emissivity cut heat transfer to $1/(N + 1)$

2023 Q3(a) 5 marks

Define view factor. Explain the crossed-strings method.

2023 Q3(b) 10 marks

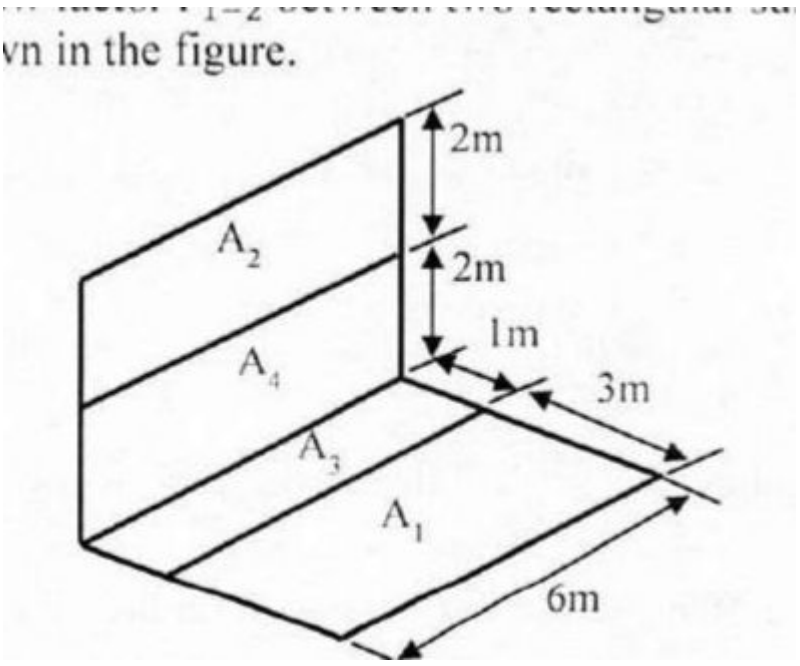
Show that N radiation shields reduce heat transfer to $1/(N + 1)$ (all emissivities equal).

2023 Q3(c) 12 marks

Equilateral triangular duct, sides 1 m: base $\varepsilon = 0.7$ at 600 K, left side black at 1000 K, right side insulated. Heat supplied to the base per metre.

2022 Q3(d) 10 marks

View factor $F_{1\text{-}2}$ between the two rectangles in the figure.



2022 Q3(a) 10 marks

Net radiation exchange between two infinite parallel plates by the network method.

2022 Q3(b) 9 marks

What is a reradiating surface? Network for a three-zone enclosure with a reradiating surface.

2021 Q3(a) 7 marks

What is view factor? State the reciprocity relation.

2021 Q3(b) 10 marks

Net radiation exchange between two infinite parallel plates by the network method.

2021 Q3(c) 5 marks

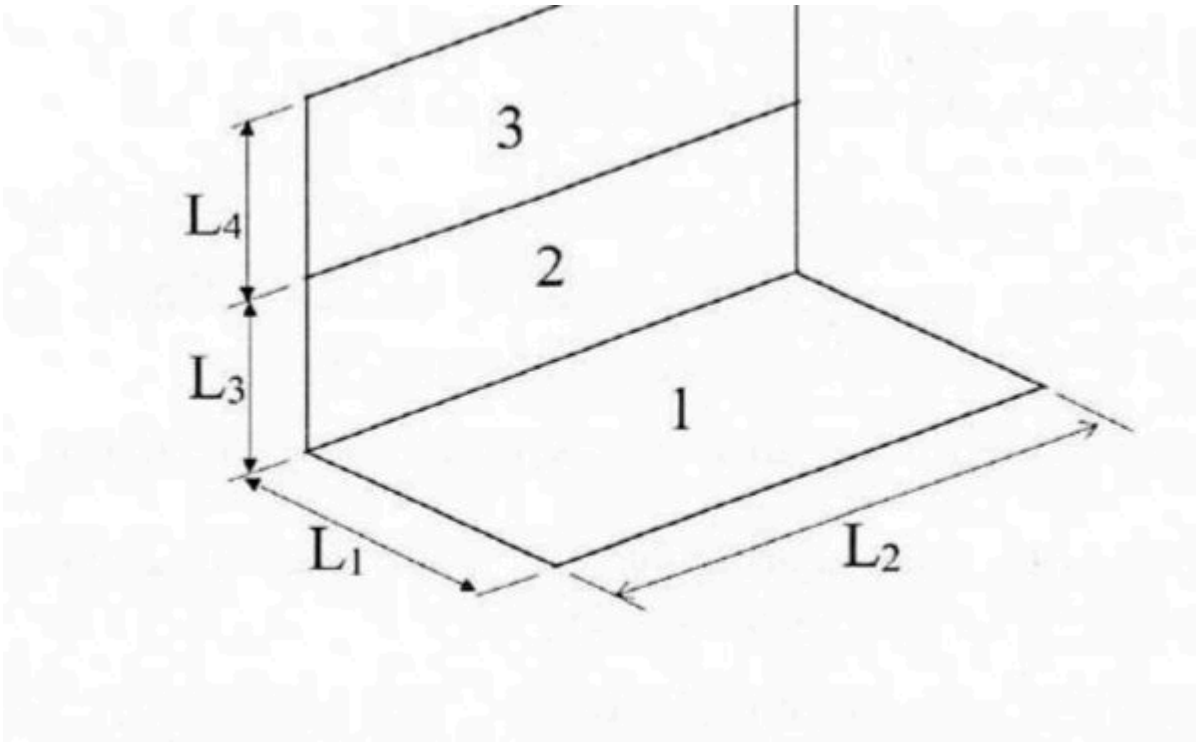
What is a radiation shield? Why is it used?

2021 Q3(d) 13 marks

Parallel plates $\varepsilon_1 = 0.7$, $\varepsilon_2 = 0.5$, 900 K and 500 K: net radiation exchange.

2020 Q2(c) 10 marks

What is view factor? Deduce the expression for $F_{1\text{-}3}$ in the figure.



2020 Q3(a) 10 marks

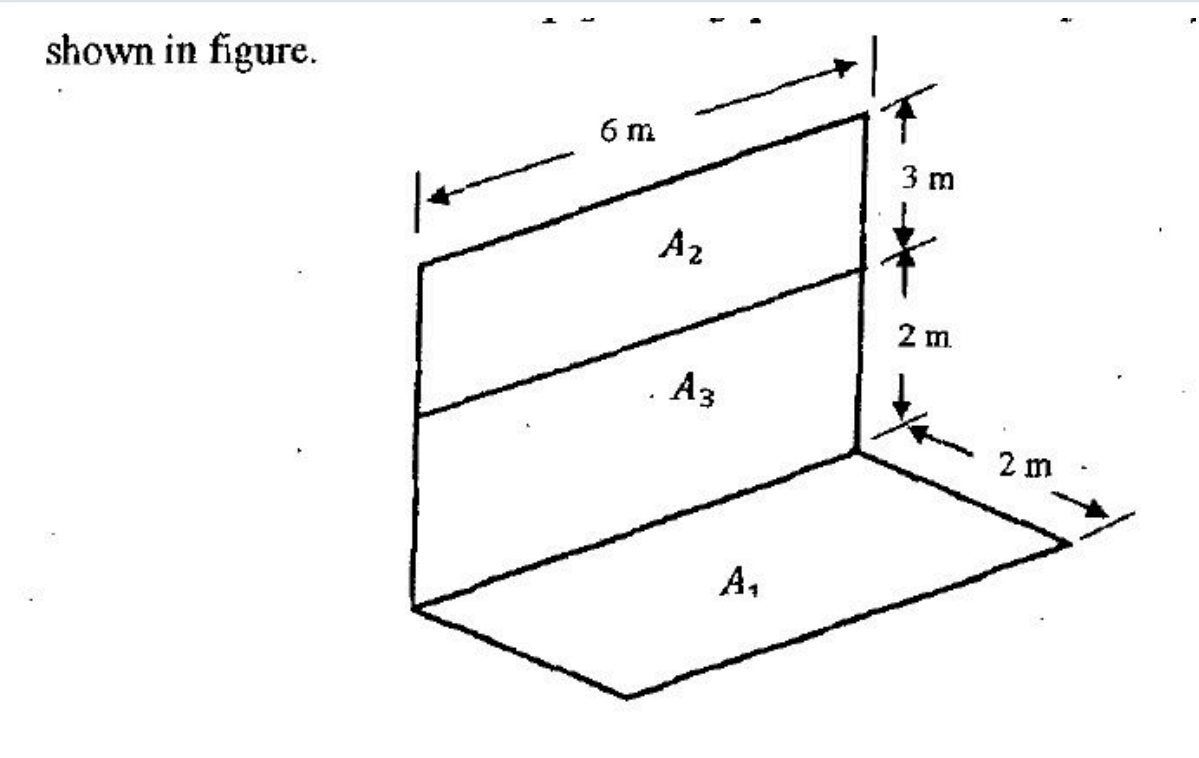
What is radiosity? Radiation network for a three-zone enclosure with a reradiating surface.

2019 Q3(a) 9 marks

View factor; reciprocity and summation relations.

2019 Q3(d) 9 marks

View factors $F_{1\text{-}2}$ and $F_{2\text{-}1}$ for the perpendicular rectangles in the figure.



2019 Q3(c) 8 marks

Net heat flow in a two-zone enclosure.

Full answer: ESE 3107 Study Pack, topic A4.

Section B: Collector performance and applications

questions 5-8

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

B4 Other applications: drying, desalination (solar still), cooling, pool vs pond, solar thermal power

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 204 marks in total

STUDY THESE

- Crop drying: open sun drying problems; direct (cabinet), indirect (separate air collector), mixed-mode; PV-powered fans
- Solar still: basin, black liner, sloping glass, collection trough; Dunkle $q_{ew} = 0.016273 h_{cw}(P_w - P_g)$, $\dot{m} = 3600 q_{ew}/\lambda$
- Solar cooling: absorption (LiBr-water), solid and liquid desiccant cooling
- Power: parabolic trough, central receiver (heliostats + tower), dish-Stirling, solar pond (salt-gradient); pool heating vs pond

2023 Q7(c) 10 marks

Advantages of solar crop drying; working principle with schematic.

2023 Q8(d) 9 marks

Basic construction of a PV crop-drying system.

2023 Q8(a) 11 marks

Explain any one solar thermal process for cooling.

2023 Q8(b) 6 marks

Differences between solar pool heating and solar pond heating.

2023 Q8(c) 9 marks

Central receiver system to generate electricity.

2023 Q7(d) 7 marks

Briefly explain solar distillation.

2022 Q8(a) 13 marks

Solar crop drying; indirect crop-drying system with schematic.

2022 Q8(b) 10 marks

Solar-assisted liquid desiccant cooling system.

2022 Q8(c) 12 marks

Solar pond and solar thermal power; solar power tower with schematic.

2021 Q7(c) 5 marks

How to overcome limitations of open sun drying.

2021 Q7(d) 12 marks

Solar-assisted desiccant cooling system.

2021 Q8(a) 8 marks

Differences between solar pool heating and solar pond.

2021 Q8(b) 14 marks

How collectors generate electricity; parabolic dish system.

2021 Q8(c) 13 marks

What is solar desalination? Still with water 50 °C, glass 40 °C, $h_{cw} = 2.40$: hourly distillate per m^2.

2020 Q6(a) 12 marks

Solar pool heating; direct solar drying system with sketch.

2020 Q6(b) 8 marks

Central receiver system for thermal power generation.

2020 Q6(c) 10 marks

Still with water 60 °C, glass 50 °C, C = 0.04, n = 0.35, $h_{cw} = 2.8$: evaporative heat transfer per m^2.

2019 Q8(a) 10 marks

What is a solar pond? Working principle.

2019 Q8(b) 13 marks

Collectors for solar thermal power; central receiver and dish systems.

2019 Q8(c) 12 marks

What is distillation? Solar distillation with diagram.

Full answer: ESE 3107 Study Pack, topic B4.

B2 Concentrating collectors: definitions, classification, maximum concentration, tracking, materials

Pass set · Very likely · 100% · Easy

Asked in 5 of 5 papers · 123 marks in total

STUDY THESE

- Aperture area, absorber (receiver) area, concentration ratio $C = A_a/A_r$, acceptance angle $2\theta_c$, intercept factor γ
- Classification: line focus (parabolic trough, linear Fresnel, CPC) and point focus (dish, central receiver/heliostats, Fresnel lens); C vs receiver temperature
- $C_{max} = 1/\sin\theta_s$ (2-D) and $1/\sin^2\theta_s$ (3-D) = 46,000 for $\theta_s \approx 0.27°$
- Tracking: single axis (E-W, N-S, polar) and two-axis; reflectors (silvered glass, aluminium), receivers, covers; advantages and disadvantages

> 2023 Q5(c) 10 marks
>
> With figure, classify concentrating collectors.

> 2023 Q5(d) 5 marks
>
> Advantages and disadvantages of concentrating collectors.

> 2023 Q6(c) 13 marks
>
> Why is tracking essential? Types of tracking for concentrators.

> 2022 Q6(b) 10 marks
>
> Define: aperture area, acceptance angle, concentration ratio, intercept factor.

> 2022 Q6(c) 10 marks
>
> Concentration ratio vs receiver temperature; practical ranges.

> 2022 Q6(d) 5 marks
>
> Characteristics a reflector should possess.

> 2021 Q6(b) 14 marks
>
> With figures, classify solar concentrators.

> 2021 Q6(c) 11 marks
>
> Why is tracking necessary? Advantages and disadvantages of concentrating collectors.

> 2020 Q5(a) 10 marks
>
> What is solar tracking? Distinguish between “three-axis” and two-axis tracking.

> 2019 Q6(a) 8 marks
>
> Define: aperture area, absorber area, acceptance angle, concentration ratio.

> 2019 Q6(d) 11 marks
>
> Show that C_{max} for a circular concentrator is 46,747.

> 2019 Q6(b) 8 marks
>
> Advantages and disadvantages of concentrating collectors.

> 2019 Q6(c) 8 marks
>
> Why is tracking needed? Materials used in concentrators.

Full answer: ESE 3107 Study Pack, topic B2.

B3 Solar water heating and storage: thermosiphon, ICS, direct/indirect, heat pump, storage tank numericals

Pass set · Very likely · 100% · Medium

Asked in 5 of 5 papers · 126 marks in total

STUDY THESE

- Passive (thermosiphon, ICS) vs active (direct/open loop, indirect/closed loop with heat exchanger, drain-back, air systems)
- Thermosiphon: hot water rises to the tank above the collector; reverse circulation at night prevented by tank 30 cm above collector top / check valve
- ICS: tank is the collector; high night losses (solutions: selective coating, transparent insulation, night cover)
- Fully mixed tank: $T_s^+ = T_s + (\Delta t/mc_p)[Q_u - L - UA(T_s - T_a)]$

> 2023 Q7(a) 10 marks
>
> Working principle of a direct solar water heating system.

> 2023 Q7(b) 8 marks
>
> Problems of ICS and possible solutions.

> 2022 Q7(b) 8 marks
>
> Advantages and disadvantages of passive solar water heating.

> 2022 Q7(c) 5 marks
>
> How to prevent reverse circulation in passive systems.

> 2022 Q7(a) 12 marks
>
> Solar heat pump systems for water heating and space heating.

> 2022 Q7(d) 10 marks
>
> 400 kg tank, UA = 10 W/°C, room 16 °C, 8 h from 7 AM, given Q_u and loads, initial 40 °C: final temperature.

> 2021 Q7(a) 10 marks
>
> Working principle of an indirect solar water heating system.

> 2021 Q7(b) 8 marks
>
> Integrated collector storage system; disadvantages.

> 2020 Q5(b) 12 marks
>
> Classify solar water heating systems; thermosiphon system with sketch.

> 2020 Q5(c) 8 marks
>
> How does a storage system work in a solar thermal system?

> 2019 Q7(a) 7 marks
>
> What is solar water heating? Applications.

> 2019 Q7(b) 10 marks
>
> Thermosiphon system with schematic.

> 2019 Q7(c) 18 marks
>
> Why is storage necessary? 500 kg tank, UA = 12 W/°C, room 20 °C, 10 h from 5 AM, given Q_u and loads, initial 45 °C: final temperature.

Full answer: ESE 3107 Study Pack, topic B3.

B1 Flat plate collector performance: losses, U_L, F', F_R, critical radiation level, testing, numericals

Next to add · Very likely · 100% · Hard

Asked in 5 of 5 papers · 184 marks in total

STUDY THESE

- Energy balance $Q_u = A_c[S - U_L(T_p - T_a)] = A_cF_R[S - U_L(T_i - T_a)]$; $S = (\tau\alpha)I_T$; $\eta = Q_u/(A_cI_T)$

- Losses: top (convection + radiation through glazing), bottom (conduction through insulation), edge; $U_L = U_t + U_b + U_e$; thermal network
- Fin efficiency $F = \tanh[m(W - D)/2]/[m(W - D)/2]$; $F' = (1/U_L)/\{W[1/(U_L(D + (W - D)F)) + 1/(\pi D_i h_{fi})]\}$; $F_R = F'F''$
- Critical radiation level $I_c = U_L(T_i - T_a)/(\tau\alpha)$; effective $(\tau\alpha)_e \approx 1.01\text{-}1.02\tau\alpha$; optimum tilt ≈ latitude

> `2023 Q5(a) 8 marks`
>
> Define: critical radiation level, collector efficiency factor, heat removal factor, semi-acceptance angle.

> `2023 Q5(b) 12 marks`
>
> With figures, significance of concentration ratio and effective (τα).

> `2023 Q6(a) 12 marks`
>
> Discuss all the losses of a flat plate collector.

> `2022 Q5(a) 8 marks`
>
> Significance of: critical radiation level, effective (τα), heat removal factor, inclination angle.

> `2022 Q5(b) 15 marks`
>
> Derive a general expression for the collector efficiency factor.

> `2022 Q6(a) 10 marks`
>
> Testing procedure for a new FPC.

> `2022 Q5(c) 12 marks`
>
> 5 m^2 FPC tested in the dark: water 65 → 50 °C at 0.05 L/s, ambient 8 °C: U_L.

> `2021 Q6(a) 10 marks`
>
> Define: acceptance angle, concentration ratio, critical radiation level, collector efficiency factor, heat removal factor.

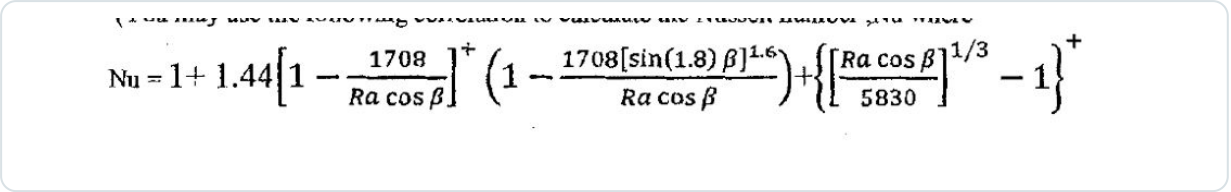
> `2021 Q5(a) 10 marks`
>
> Thermal network for a double-glass-covered FPC.

> `2021 Q5(c) 19 marks`
>
> U_L for an FPC at 45°: T_a = 20 °C, glass 50 °C, plate 70 °C, 5 cm glass wool (k = 0.03), air gap “75 cm”, ε_g = 0.8, ε_p = 0.95.

> `2020 Q4(a) 6 marks`
>
> Define: concentration ratio, critical radiation level, collector efficiency factor.

> **2020 Q4(c) 16 marks**
>
> FPC 4 m^2, 0.06 kg/s, U_L = 6.9, W = 120 mm, D = 15/13.5 mm, plate 0.4 mm copper, h_{fi} = 320, τα = 0.8, 2.88 MJ/m^2 in 1 h, $T_i - T_a$ = 5 °C: F, F′, Q_u, η.

> **2019 Q4(b) 16 marks**
>
> Define and explain: critical radiation level, collector efficiency factor, effective (τα), heat removal factor.

> **2019 Q5(b) 30 marks**
>
> U_L for an 8 m × 2.5 m FPC at 45° (plate 69 °C, air 20 °C, gap 75 mm, ε_p = 0.95, ε_g = 0.88, wind 2.5 m/s, back 60 mm and edge 20 mm insulation k = 0.04, depth 0.08 m).
>
> 
> $$Nu = 1 + 1.44\left[1 - \frac{1708}{Ra\cos\beta}\right]^{+}\left(1 - \frac{1708[\sin(1.8)\beta]^{1.6}}{Ra\cos\beta}\right) + \left\{\left[\frac{Ra\cos\beta}{5830}\right]^{1/3} - 1\right\}^{+}$$

Full answer: ESE 3107 Study Pack, topic B1.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
CPC	compound parabolic collector
CSP	concentrated solar power
DNI	direct normal irradiance
ETC	evacuated tube collector
FPC	flat plate collector
GHI	global horizontal irradiance
ICS	integrated collector storage
kWh	kilowatt-hour
LiBr	lithium bromide
LST	local standard time
MJ	megajoule

PV	photovoltaic
sr	steradian
ST	solar time
WMO	World Meteorological Organization

Built from the scanned ESE 3107 papers (2019, 2020, 2021, 2022, 2023). Question text is cleaned up from the scans.